

Formally Verified Delta-Hedging: A Lean 4 Accounting Kernel with Empirical Validation Across Four Historical Regimes

backtest-proofs — eigenq-xyz research series

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TODO: Draft last (§1 per fill order). 150–250 words. State the question, the zero-sorry claim at the commit SHA below, the WRDS OptionMetrics 2019–2024 sample, the headline MC deviation percentage, and the scope limitation (flat- Black-Scholes; no transaction costs).

i PLAN: strategic context (strip before publish)

Four credibility levers from PLAN-backtest-credibility.md: 1. P&L attribution identity — §9 Discussion only (Carr-Madan scatter covers gamma attribution) 2. Leland (1985) rehedg-frequency sweep — Figure 5 (§8 Robustness) 3. QuantLib A-B comparison — Figure 6 (§7 Results) 4. Stress runs across four historical regimes — Figure 7 (§8 Robustness)
Authoring fill order: §5 → §6 → §7 → §8 → §3 → §4 → §2 → §1. Each pass replaces one section’s TODO with real prose.

i PLAN: formalization table (confirmed sorry-free at v0.4)

Paper theorem	Lean name	File	Line
NAV identity	valueIdentity	BacktestProofs/Invariant	311
Trade accounting	valueUpdateFormula	BacktestProofs/Invariant	212
Cash update	cashUpdateCorrect	BacktestProofs/Invariant	32
Quantity conservation	quantityConservation	BacktestProofs/Invariant	176
Self-financing	selfFinancing	BacktestProofs/Invariant	381
Well-formedness	applyTrade_wellFormed	BacktestProofs/Invariant	362
Settlement (crown jewel)	settlement_value_forBacktest	BacktestProofs/SettlementInvariant	104
Call payoff 0	callPayoff_nonneg	quant-core/lean/QuantCore/OptionInvariants	32

```
Put payoff 0      putPayoff_nonneg    quant-core/lean/QuantCore/OptionInvariants.lean
Option payoff 0   optionPayoff_nonneg quant-core/lean/QuantCore/OptionInvariants.lean
```

All 10 theorems confirmed sorry-free at main HEAD.

Abstract

TODO: Draft last. See hidden callout above for content requirements.

1 Introduction

Note

TODO (§2, draft second-to-last): 1. **Why now:** AI-generated quant code needs formal accounting verification. 2. **What we do:** 10 Lean theorems + 7 empirical figures (Hull, MC, BKL, Carr-Madan, QuantLib A-B, Leland sweep, four stress regimes). 3. **What is new:** Only system combining Lean-verified accounting with real WRDS stress-period validation across four distinct crisis regimes. 4. **Roadmap:** one sentence per section.

TODO: Draft per fill order (§2 = second-to-last).

2 Setup

Note

TODO (§3, draft fifth): - Discrete-time delta-hedging model - **Portfolio, Position, Trade** primitives from **BacktestProofs.Basic** - Basis-point integer representation ($\times 10,000$); why integers, not floats - BS delta strategy: **SingleLegStrategy** protocol - Reference Lean files by path

TODO: Draft per fill order (§3 = fifth).

3 Main Results

Note

TODO (§4, draft fourth): 10 informal+formal theorem pairs. Lead each with intuition; follow with the Lean statement. Use `::: {#thm-name}` callout format. Every theorem references its Lean name and file from the table in §5.

TODO: Draft per fill order (§4 = fourth).

4 Formalization

Note

TODO (§5, draft first — establishes what is and isn't proved). The content below is the live table; fill prose around it.

4.1 Lean 4 artifacts

The formal development lives in [backtest-proofs/lean/](#). The following table maps paper-level theorems to their Lean 4 names and source files.

Table 4.1: Lean 4 theorem map. All 10 theorems are machine-checked with zero `sorry`.

Paper theorem	Lean name	Source file
NAV identity	<code>valueIdentity</code>	<code>BacktestProofs/Invariants.lean</code>
Trade accounting	<code>valueUpdateFormula</code>	<code>BacktestProofs/Invariants.lean</code>
Cash update	<code>cashUpdateCorrect</code>	<code>BacktestProofs/Invariants.lean</code>
Quantity conservation	<code>quantityConservation</code>	<code>BacktestProofs/Invariants.lean</code>
Self-financing	<code>selfFinancing</code>	<code>BacktestProofs/Invariants.lean</code>
Well-formedness	<code>applyTrade_wellFormed</code>	<code>BacktestProofs/Invariants.lean</code>
Settlement (crown jewel)	<code>settlement_value_formula</code>	<code>BacktestProofs/SettlementInvariants.lean</code>
Call payoff 0	<code>callPayoff_nonneg</code>	<code>quant-core/QuantCore/OptionInvariants.lean</code>
Put payoff 0	<code>putPayoff_nonneg</code>	<code>quant-core/QuantCore/OptionInvariants.lean</code>
Option payoff 0	<code>optionPayoff_nonneg</code>	<code>quant-core/QuantCore/OptionInvariants.lean</code>

Formalization claim

At commit `{python} COMMIT_SHA`, `lake build` succeeds for `backtest-proofs` and `grep -rn sorry --include="*.lean"` returns no matches in the proof tree. Reproduce with `make`

verify-lean or the verify-lean skill.

4.2 Scope of formalization

TODO: Draft what is and is not formalized (accounting layer only; not pricer floating-point, FFI bridge, or statistical claims).

5 Empirical Setup

Note

TODO (§6, draft second): Two data tables: (a) WRDS OptionMetrics 2019–2024 main sample, (b) four stress-period downloads. GBM simulator and Hull reference paths documented separately. Per the source-financial-data skill: data source, sample window, universe, filters, N observations, licensed-data note (never committed).

Use the data table format from `_template.qmd`.

5.0.1 Main sample

TODO: WRDS OptionMetrics 2019-01-01 to 2024-12-31. Tickers: SPY, QQQ, AAPL, MSFT, JPM. ATM $\pm 3\%$ filter; 20–40 day expiry filter. SOFR rate: 1.65% annualised average. Data not committed; download via `backtest-proofs/docs/wrds_data_download.md`.

5.0.2 Stress-period samples

TODO: Four supplementary WRDS downloads, one per crisis window. Same schema as main sample.

5.0.3 Synthetic GBM and deterministic paths

The Monte Carlo convergence, BKL scaling, Carr-Madan, Leland, and QuantLib figures use no licensed data. GBM paths are seeded with `{python} MC_SEED` for full reproducibility.

Hull Tables 19.2 and 19.3 use the hardcoded weekly prices from Hull (2014) (Tables 19.2/19.3): $S = \$49$, $K = \$50$, $r = 5\%$, $\sigma = 20\%$, $T = 20$ weeks.

6 Results

Note

TODO (§7, draft third): Lead with MC convergence (headline). Then BKL, Carr-Madan, Hull, QuantLib A-B, WRDS holdout. One paragraph per figure: state the result, cite the figure, note the key number from metrics.json.

6.1 Monte Carlo convergence

Across `{python} f"{MC_N_PATHS:}"` seeded GBM paths the running-mean hedging cost converges to the Black-Scholes price with a final deviation of `{python} f"{mc_mean_deviation_pct:.2f}"%` ([?@fig-mc](#)).

6.2 BKL variance scaling

The empirical ratio $\text{std}(N=10) / \text{std}(N=20) = \text{{python} f"{bkl_ratio_10_20:.3f}"}$ (theory: $\sqrt{2}$ 1.414). The BKL line fits well across all five frequencies ([?@fig-bkl](#)).

6.3 Carr-Madan gamma attribution

Hedging cost correlates with gamma P&L at $r = \text{{python} f"{carr_madan_corr:.3f}"}$ across `{python} CM_N_PATHS` paths, consistent with the Carr-Madan decomposition [Carr and Madan \(1998\)](#) ([?@fig-cm](#)).

6.4 Hull Table 19.2 / 19.3 regression

For Hull Table 19.2 our hedging cost is $\text{{python} f"{hull_192_cost:,.0f}"}$ vs the reference `python f"HULL192EXPECTEDCOST :,.0f".ForTable19.3 :``{python} f"{hull_193_cost:,.0f}"` vs $\text{{python} f"{HULL_193_EXPECTED_COST:,.0f}"}$. All step certificates pass for both paths ([?@fig-hull](#)).

6.5 QuantLib A-B comparison

Note

TODO (§7 QuantLib A-B prose): Report `ql_max_dev` and `ql_within_1bp` from the chunk above. If `ql_available` is False, note that QuantLib is not installed and defer to the replication instructions in the appendix.

TODO: Draft prose after running with QuantLib installed (`uv sync --group research`).

6.6 WRDS 2019–2024 holdout

Note

TODO (§7 WRDS holdout prose): Report `wrds_holdout_median` and `wrds_holdout_n`. If `WRDS_PRESENT` is False, note the WRDS download requirement and that stress figures will be added after data arrives.

TODO: Draft after WRDS parquets are downloaded.

7 Robustness

Note

TODO (§8, draft third-ish, alongside §7): Two robustness dimensions required: 1. Sample-period sensitivity: the four stress regimes (COVID, GFC, Volmageddon, Flash Crash) — each is a distinct stressor, not cherry-picking. 2. Specification sensitivity: Leland (1985) rehedg-fre-quency sweep — does the BKL result hold at other frequencies? 3. Data source A-B: GBM vs WRDS for MC convergence (if WRDS present).

7.1 Four historical stress regimes

Note

TODO (§8 stress prose): Describe each regime, its stressor, and whether cost/premium is above/below 1.0 and why. State that all certificates pass. Reference the Lean settlement theorem (`settlement_value_formula`) as the formal guarantee that accounting invariants hold regardless of vol level.

TODO: Draft after WRDS parquets are downloaded.

7.2 Leland (1985) rehedg-fre-quency sweep

TODO: Draft prose on what the slope ratio says and how it compares to Leland (1985) Table II.

8 Discussion

Note

TODO (§9, draft sixth/last — after all results): 1. What formal proofs guarantee vs empirics: the Lean kernel proves accounting invariants hold by construction, regardless of the price path. It does NOT prove convergence to BS price (that is an empirical finding, not a theorem). 2. Flat- BS limitation: implied vol varies across the volatility surface; we use inception-date per-series implied vol as (no look-ahead). This understates variance risk during regime changes. 3. Extensions: Carr-Wu (2009) realized-vs-implied vol direction test; mortgage-proofs analogue (multi-agent routing invariants); Leland transaction-cost modification.

TODO: Draft last.

9 Appendix

9.1 A. Replication

```
git clone https://github.com/eigenq-xyz/quant-proofs
cd quant-proofs/backtest-proofs

# 1. Verify Lean proofs (Levels 1-2)
make verify-lean

# 2. Run Python tests (Levels 2-4)
make test

# 3. Install research dependencies and generate figures
cd python && uv sync --group research
cd ..
make paper
```

After downloading WRDS data per docs/wrds_data_download.md, re-run `make paper` to regenerate `?@fig-stress` with real market data.

9.2 B. Metrics snapshot

References

Carr, Peter, and Dilip Madan, 1998, Towards a theory of volatility trading, *Volatility: New estimation techniques for pricing derivatives*, 417–427.

Hull, John C., 2014, *Options, Futures, and Other Derivatives*. 9th Global. (Pearson).